

BOW ISLAND IRRIGATION AGDM, AB, Canada

WMO: 715190

Lat: **49.8684N**Lon: **111.3664W**Elev: **792**StdP: **92.17**Time Zone: **-7.00 (NAM)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.7	-24.5	-31.0	0.2	-27.5	-27.7	0.3	-24.4	10.7	-4.0	9.6	-5.8	2.4	190	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	32.1	17.2	30.2	16.7	28.3	16.2	19.3	27.6	18.2	26.8	17.2	25.8	3.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.5	12.9	22.8	15.1	11.8	21.5	14.0	11.0	20.4	58.5	27.5	54.7	26.9	51.4	25.8	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.4	-33.5	35.5	3.6	1.7	-36.1	36.7	-38.2	37.7	-40.2	38.6	-42.8	39.8		
(5)			-33.7	21.7	3.6	1.6	-36.2	22.8	-38.3	23.7	-40.3	24.6	-42.9	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-6.3	-7.6	-0.3	6.0	11.9	16.1	19.6	18.3	13.3	6.4	-1.0	-7.3
	DBStd	11.44	8.23	8.60	7.59	5.04	4.47	3.36	2.97	3.47	4.67	5.14	7.54	8.29
	HDD10.0	2505	505	493	320	134	31	1	0	0	19	133	331	537
	HDD18.3	4696	763	726	577	369	203	84	21	44	162	371	578	795
	CDD10.0	984	0	0	1	16	90	183	296	257	118	21	2	0
	CDD18.3	134	0	0	0	0	4	17	59	43	11	0	0	0
	CDH23.3	2191	0	0	0	10	126	255	862	695	234	8	0	0
	CDH26.7	761	0	0	0	1	28	66	324	262	79	0	0	0
Wind	WSAvg	3.4	3.8	3.5	3.8	3.9	3.7	3.3	2.8	2.8	3.1	3.5	3.6	3.6
Precipitation	PrecAvg	364	17	13	19	30	47	83	36	34	31	21	15	14
	PrecMax	532	44	29	50	68	102	207	125	80	68	70	32	32
	PrecMin	193	6	2	1	7	7	15	3	2	0	6	2	0
	PrecStd	76	10	7	12	15	24	41	28	21	20	14	9	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	11.0	18.3	24.6	29.7	31.7	34.5	33.7	32.4	24.4	17.8	10.7
		MCWB	5.4	4.8	8.9	11.1	14.6	17.8	17.9	17.0	15.7	12.8	8.9	4.9
	2%	DB	7.7	7.5	15.2	20.8	26.8	28.5	32.3	31.8	29.0	20.4	13.5	7.2
		MCWB	3.5	3.4	7.4	9.9	13.5	16.4	18.3	16.4	14.8	11.1	7.0	3.4
	5%	DB	5.8	5.4	12.4	18.0	24.1	26.3	30.2	29.7	26.0	17.3	10.8	5.4
		MCWB	2.6	2.1	6.0	8.4	12.4	15.6	17.8	16.2	13.9	9.6	5.7	2.3
	10%	DB	4.3	3.6	9.6	15.2	21.4	24.1	28.2	27.5	22.9	14.8	8.4	3.7
		MCWB	1.7	1.0	4.4	6.9	11.2	14.5	17.1	15.8	12.7	8.2	3.9	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	6.0	9.4	11.9	15.3	19.7	21.8	19.4	17.3	13.5	10.0	5.7
		MCDB	10.1	10.1	16.6	22.1	27.2	26.7	29.3	28.3	26.4	22.4	16.6	9.6
	2%	WB	3.9	3.7	8.0	10.4	14.0	17.8	20.1	18.2	15.7	11.6	7.6	3.7
		MCDB	7.3	7.1	14.1	20.1	24.7	26.3	27.9	27.5	26.7	18.9	13.1	6.5
	5%	WB	2.7	2.2	6.3	8.8	13.0	16.4	18.9	17.2	14.4	10.1	5.9	2.4
		MCDB	5.5	5.0	11.9	16.7	22.3	24.3	27.4	26.6	24.5	16.2	10.3	5.4
	10%	WB	1.7	1.0	4.5	7.5	12.0	15.3	17.9	16.4	13.2	8.7	4.4	1.0
		MCDB	4.3	3.5	9.2	14.3	19.7	22.4	26.1	25.3	22.3	14.4	8.1	3.5
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.6	11.0	12.6	13.2	12.6	15.0	15.1	14.3	12.3	10.8	9.6
		MCDBR	10.1	12.1	15.0	18.2	18.0	16.2	18.5	19.3	19.9	16.0	13.6	10.3
		MCWBR	7.2	8.8	8.6	9.1	7.4	7.0	7.7	7.3	8.1	8.2	8.3	7.3
	5% WB	MCDBR	10.1	11.7	14.4	16.6	15.8	15.0	16.6	16.7	18.9	15.0	13.2	10.7
		MCWBR	7.4	8.7	8.6	8.6	6.9	7.2	7.6	7.0	8.0	8.0	8.5	8.0
Clear-Sky Solar Irradiance	taub	0.268	0.271	0.295	0.346	0.368	0.359	0.358	0.380	0.337	0.295	0.265	0.257	
	taud	2.416	2.471	2.463	2.315	2.299	2.380	2.397	2.306	2.407	2.506	2.493	2.411	
	Ebn at Noon	796	892	923	898	889	899	893	853	860	848	799	757	
	Edh at Noon	61	74	90	116	124	115	112	117	95	71	56	53	
All-Sky Solar Radiation	RadAvg	1.25	2.24	3.57	4.94	5.85	6.41	6.97	5.61	4.00	2.49	1.39	0.97	
	RadStd	0.12	0.19	0.25	0.34	0.42	0.38	0.35	0.35	0.33	0.25	0.10	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page